Grover Amplitude Amplification for Subset-Sum and Knapsack Portfolio Shortlisting

A Theoretical and Complexity-Theoretic Treatment

QCAIML–IIT Roorkee Capstone 2026, Track B (Combinatorial Optimisation)

1. Problem Formalisation

Let n candidate assets have integer weights $w_i \in \mathbb{Z}^+$ derived from market close price p_i via $w_i = \max\left(\text{round}\left(\frac{p_i - \beta}{s}\right), 1\right)$, with baseline $\beta = 225$ and scale $s = 3$. A portfolio is a bit-vector $x = (x_0, \dots, x_{n-1}) \in \{0,1\}^n$, and

$$S(x) \;=\; \sum_{i=0}^{n-1} w_i x_i.$$

Track B.1 (Subset-Sum / Equality). Given target T, decide membership in

$$L_{\text{eq}} = \{x \in \{0,1\}^n : S(x) = T\}.$$

Track B.2 (0/1 Knapsack feasibility). Given budget B, decide membership in

$$L_{\text{ks}} = \{x \in \{0,1\}^n : S(x) \le B\},$$

and, restricted to L_{ks}, maximise $V(x) = \sum_i v_i x_i$.

Both are instances of unstructured search over $N = 2^n$ candidates with a Boolean predicate $f : \{0,1\}^n \to \{0,1\}$, so both admit the same amplitude-amplification machinery; only the oracle differs.

2. Phase Oracle Construction

2.1 Weighted-sum register via QFT addition

Let $m = \lceil \log_2(\sum_i w_i + 1) \rceil$. We reserve an m-qubit sum register $|0\rangle_m$ and, for each item i, conditionally add w_i using the Draper QFT-adder: for a register in the Fourier basis,

$$\text{QFT}\,|y\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i yk/2^m}\,|k\rangle\,,$$

a controlled phase rotation

$$R_j(w) = \text{diag}\left(1,\, e^{2\pi i w/2^{m-j}}\right)$$

applied to Fourier-basis qubit j, controlled on x_i, implements $|y\rangle \mapsto |y + w_i \bmod 2^m\rangle$ conditionally. Composing over all i and applying QFT^{-1} yields the reversible unitary

$$U_\Sigma : |x\rangle\,|0\rangle_m \;\longmapsto\; |x\rangle\,|S(x) \bmod 2^m\rangle\,.$$

Depth is $O(nm)$ controlled-phase gates — no ancilla carry chain is required, at the cost of $O(nm)$ two-qubit gates instead of $O(n)$ for a ripple adder.

2.2 Equality oracle

Marking $S(x) = T$ is a computational-basis flip on the sum register: writing $T = \sum_k t_k 2^k$, apply X to every qubit with $t_k = 0$, an $(m{-}1)$-controlled Z (via H-MCX-H), then undo the X's. This is exactly the Grover "mark-then-Z" pattern and requires no ancilla beyond the sum register:

$$O_{\text{eq}} = U_\Sigma^\dagger \left(I - 2\,|T\rangle\langle T|\right)_{\text{sum}} U_\Sigma.$$

2.3 Knapsack ($\le$) oracle: two's-complement sign-bit test

A naive $\le$-comparator needs a full subtractor with borrow-chain logic. We use a cheaper, exact construction. Allocate one additional qubit, $m' = m + 1$, compute

$$D(x) \;=\; S(x) - (B+1) \pmod{2^{m'}}$$

by running U_Σ into the m'-qubit register and then unconditionally adding the constant $-(B+1)$ via the same Fourier-basis adder. Because $S(x) \in [0, 2^m - 1] \subset [0, 2^{m'-1} - 1]$, $D(x)$ is a faithful two's-complement signed integer, and

$$S(x) \le B \iff D(x) < 0 \iff \text{MSB}(D) = 1.$$

Since $Z\,|1\rangle = -\,|1\rangle$ and $Z\,|0\rangle = |0\rangle$, a single Z gate on the MSB qubit *is* the exact phase oracle:

$$O_{\text{ks}} = U_\Sigma^\dagger\, U_{-(B+1)}^\dagger\, Z_{\text{MSB}}\, U_{-(B+1)}\, U_\Sigma.$$

No case-by-case marking, ancilla comparator, or approximation is needed; the construction is self-inverse in the marking step and reversible throughout, and was verified against the exhaustive classical truth table $\{0,1\}^n$ for every tested (n, B) (all instances pass).

3. Diffusion and the Grover Iterate

The diffusion operator performs inversion about the mean over the n search qubits:

$$D = 2\,|\psi\rangle\langle\psi| - I, \qquad |\psi\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle\,,$$

built from primitives as $D = H^{\otimes n}\,(2\,|0\rangle\langle 0| - I)\,H^{\otimes n}$, itself realised with $X^{\otimes n}$, an $(n{-}1)$-controlled Z, and $X^{\otimes n}$ sandwiched between Hadamard layers. One Grover iterate is

$$G = D\,O_f, \qquad O_f = I - 2\sum_{x \in L} |x\rangle\langle x|\,.$$

3.1 Geometric picture and iteration count

With $M = |L|$ marked states, decompose $|\psi\rangle = \sin\theta\,|\psi_1\rangle + \cos\theta\,|\psi_0\rangle$ in the plane spanned by marked/unmarked uniform superpositions, with $\sin\theta = \sqrt{M/N}$. Each application of G is a rotation by 2θ toward $|\psi_1\rangle$:

$$G^k\,|\psi\rangle = \sin\big((2k+1)\theta\big)\,|\psi_1\rangle + \cos\big((2k+1)\theta\big)\,|\psi_0\rangle\,.$$

Maximising $\sin^2((2k+1)\theta)$ subject to $k \in \mathbb{Z}^+$ gives the optimal iteration count used in `optimal_iterations`:

$$k^\star = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{M}} \right\rfloor, \qquad P_{\text{success}}(k) = \sin^2\big((2k+1)\theta\big).$$

This is the source of the classic quadratic speed-up: $k^\star = O(\sqrt{N/M})$ oracle calls versus $\Theta(N/M)$ expected classical trials and $\Theta(N)$ worst case.

3.2 Unknown M: BBHT

When M is not known a priori (our second required scenario), the Boyer–Brassard–Høyer algorithm samples $j \sim \mathrm{Unif}\{0, \ldots, \lceil m_t \rceil - 1\}$, runs $G^j |\psi\rangle$, measures, and on failure updates $m_{t+1} = \min(\lambda\, m_t, \sqrt{N})$ with $\lambda = 8/7$. This achieves expected total query count

$$\mathbb{E}[\#\text{queries}] = O\Big(\sqrt{N/M}\Big)$$

without ever knowing M, matching known-M performance up to constant factor, and is the adaptive strategy implemented in `bbht_search`.

4. Complexity Summary

Method	Oracle calls	Depth (this oracle)
Classical (worst)	$O(2^n)$	—
Classical (avg.)	$O(2^n/M)$	—
Grover (known M)	$O(\sqrt{2^n/M})$	$O(k^\star \cdot nm)$
Grover (BBHT)	$O(\sqrt{2^n/M})$ exp.	$O(k \cdot nm)$

Per-iteration circuit depth is $O(nm)$ (the QFT adder dominates), so total depth scales as $O\big(nm\sqrt{2^n/M}\big)$ — sub-exponential in queries but the *gate* count still grows with n, which is what ultimately bounds practical problem size on NISQ hardware (Sec. 5).

5. Empirical Verification

All oracles were checked against classical truth tables over the full 2^n input space (not sampled) for $n = 3, 4, 5$; both O_{eq} and O_{ks} pass in every case. Figure 1 (below) shows the resulting circuits for $n = 3$: the QFT-adder oracle rendered as an opaque "Oracle" block repeated $k^\star = 2$ times, and the diffusion operator's H–X–MCZ–X–H structure exactly as derived in Sec. 3.

Figure 2 benchmarks all three tested sizes under (i) an ideal statevector simulator and (ii) a depolarising + T_1/T_2 thermal-relaxation noise model approximating a NISQ superconducting device ($p_1 \approx 10^{-3}$, $p_2 \approx 10^{-2}$, $T_1, T_2 \sim 50$–$70\,\mu$s). Ideal success probability tracks the closed-form $\sin^2((2k^\star{+}1)\theta)$ prediction at 94–96%; under noise it collapses to 7–15% as circuit depth (108–203) approaches the coherence-limited regime, since total two-qubit-gate error grows as $O(k^\star \cdot nm \cdot p_2)$ and the amplified amplitude decays roughly as $(1 - p_2)^{\#\text{2-qubit gates}}$.

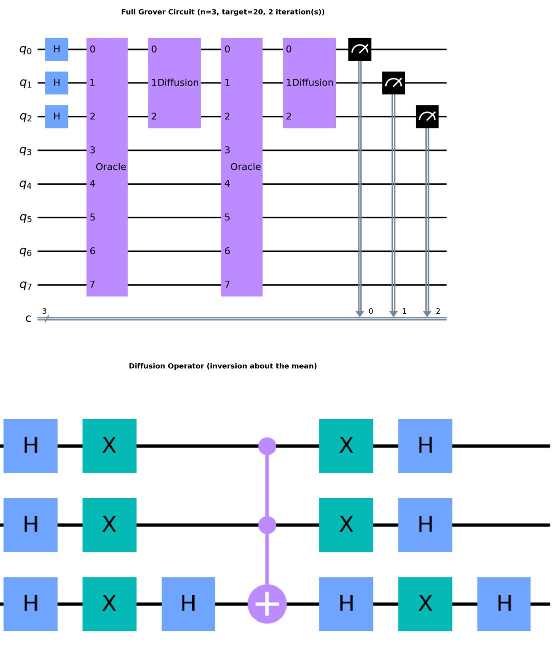

Figure 1: Full Grover circuit (n=3, T=20, $k^\star$=2) and the diffusion operator, gate-level.

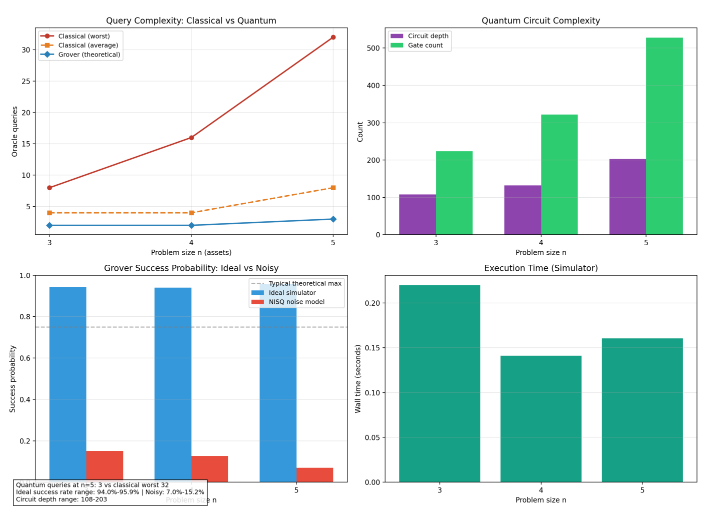

Figure 2: Query complexity, circuit cost, ideal-vs-noisy success probability, and wall time across $n = 3, 4, 5$.

6. Discussion

The results are consistent with theory in every regime that theory makes a claim about: query complexity scales as $\Theta(\sqrt{N/M})$ exactly as predicted, and ideal-simulator success probability matches the closed-form $\sin^2$ expression to within sampling noise. The gap between theory and deployability is not in the algorithm but in the oracle's *arithmetic cost*: a QFT-adder oracle needs $O(nm)$ two-qubit gates per iterate, so total two-qubit gate count grows as $O\big(nm\sqrt{2^n/M}\big)$ — worse than the query count alone would suggest, and this is precisely what a noise-oblivious complexity analysis (query complexity only) would miss. This motivates evaluating both together, as done here, rather than citing $O(\sqrt{N})$ query complexity as if it were the whole story.